Example 2

(i) Write out all the factors of 40.
(ii) Write out all the factors of 60.
(iii) Find the highest common factor of 40, 60.

(i) The factors of 40 are 1, 2, 4, 5, 8, 10, 20, 40.
(ii) The factors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.
(iii) The highest common factor is 20.

Exercise 1A

1. Using all the digits 4, 3, 1, 7 what is:
(i) the largest four–digit number that can be formed from them?
(ii) the smallest

2. What is the difference between the largest and smallest four–digit numbers that can be formed using all the digits 2, 9, 3, 5?

3. N = 1, 2, 3, 4, 5, ...
(i) List the even natural numbers from 7 to 19
(ii) List the odd natural numbers less than 13.

4. Place the following numbers in order, starting with the smallest:
(i) 58, 45, 14, 21, 67, 61, 17, 32
(ii) 345, 542, 687, 364, 194, 445, 538.

5. Write down the first five multiples of the following numbers:
(i) 3 (ii) 5 (iii) 7 (iv) 9 (v) 12

6. What is the lowest common multiple of the following pairs of numbers:
(i) 4, 6 (ii) 6, 8 (iii) 3, 7 (iv) 9, 12 (v) 10, 12

7. Write down all the prime numbers between 1 and 40.

8. Write 26 as the sum of two prime numbers.

9. Write the number 30 as:
(i) the sum of 3 prime numbers
(ii) the product of 3 prime numbers.

10. In the set of numbers 3, 4, 5, 9, 10, 12, 15, 18, 30.
(i) Which numbers are prime?
(ii) Which numbers are factors of 30?
(iii) Which numbers have 6 as a divisor?
(iv) How many numbers are divisible by 5?

11. Write the following numbers in figures:
(i) Seven thousand and twenty six
(ii) Sixteen thousand, three hundred and twenty five
(iii) One thousand and eleven.

12. Twelve thousand three hundred and fifteen fans went to a concert on Friday and eleven thousand five hundred and forty four went on Saturday.
Find the total number who went to the concert in figures.

13. The figures below show the number of people in the seats in each row of a cinema:
18, 17, 18, 12, 7, 14, 15, 10, 8, 5
The cinema holds 200 people in total.
How many empty seats are there?

14. Complete the following chart to find the lowest common multiple of 10, 12 and 15:

Multiples	1st	2nd	3rd	4th	....	....
10						
12						
15						

15. The factors of 12 are 1, 2, 3, 4, 6, 12.
Write down the factors of the following numbers:
(i) 8 (ii) 21 (iii) 24 (iv) 28 (v) 42

16. Write down all the factors of:
(i) 40 (ii) 48.
Using your answers write down the highest common factor of 40 and 48.

17. A train travelling from Dublin to Sligo stops at Mullingar, Longford, Carrick−on−Shannon and Sligo. The following chart shows the number of passengers who get on and off the train at Dublin, Mullingar, Longford and Carrick−on−Shannon.
How many passengers arrive at Sligo?

	Dublin	Mullingar	Longford	Carrick−on−Shannon	Sligo
On	136	13	18	24	0
Off	0	45	32	8	?

Without using a calculator complete the following operations: (18 – 33)

18. $5 \times 4 + 6 - 2$

19. $7 + 3 \times 9 - 5$

20. $6 \times 3 + 36 \div 4$

21. $39 \div 3 + 2 \times 5$

22. $12 + 8 \div 2 + 4 \times 3$

23. $5 \times 8 - 6 + 45 \div 3$

24. $7 + 9 \div 3 + 6 \times 2 - 4 \div 2$

25. $15 - 10 \div 2 + 6 \times 4$

26. $(36 - 12) \div 3 + 6 \times 4$

27. $25 \div (8 - 3) + 6 \times (9 - 2)$

28. $16 \div 8 + 40 \div (5 + 3)$

29. $45 \div (20 - 11) + 3 \times (8 - 2)$

30. $(9 + 7) \div 4 + 3 \times (11 - 8)$

31. $(7 + 3 \times 2) - (45 \div 9 + 6)$

32. $2 \times (7 - 4) + 28 \div 7 + 5$

33. $(16 + 4) \div 5 + (10 \times 2 - 16)$

34. $2 + 33 \div 11 + (6 - 12 \div 4)$

35. $(10 \times 3 - 20) \div (20 \div 5 - 2)$.

36. Fill in the missing operations using only $+$, $-$, $\times$, or $\div$ to complete the equation correctly:
(i) $11 \square 3 \square 3 = 20$
(ii) $10 \square 6 \square 5 = 40$
(iii) $18 \square 3 \square 17 = 23$
(iv) $(4 \square 3) \square (2 \square 5) = 49$

37. (i) Find the value of $(12 - 2) \times (3 + 1)$.
(ii) Find the value of $12 - 2 \times 3 + 1$.
(iii) Add brackets to make the equation in the box below correct.

$$12 - 2 \times 3 + 1 = 4$$